

Name _____

Teacher _____

**GOSFORD HIGH SCHOOL****2015****HIGHER SCHOOL CERTIFICATE****ASSESSMENT TASK 3****MATHEMATICS – EXTENSION 1****Time Allowed** - 60 minutes plus 5 minutes reading time

- Write using a black or blue pen. Black pen is preferred.
- Board approved calculators may be used.
- Answers to Questions 1-4 are to be done on the answer sheet provided.
- Questions 5-7 are to be answered on your own paper. Start each of Questions 5, 6 and 7 on a new page
- Relevant mathematical reasoning and/or calculations must be shown for Questions 5-7.

Multiple choice		/4
Question 5 a) and b)	Applications of Calculus to the Physical World	/9
Question 5 c) Question 6 a)	General Solutions to Trigonometric Equations	/6
Question 6 b) c) Question 7	Inverse Functions and Inverse Trigonometric Functions	/21
TOTAL		/40

Answer Questions 1 – 4 on the multiple choice answer sheet provided. Questions are worth 1 mark each.

1) $\tan^{-1}\left[\tan\frac{2\pi}{3}\right] =$

A) $\frac{\pi}{3}$

B) $\frac{-\pi}{3}$

C) $\frac{2\pi}{3}$

D) $\frac{-2\pi}{3}$

2) If $\cos\theta = \frac{1}{2}$, then

A) $\theta = n\pi \pm \frac{\pi}{3}$

B) $\theta = n\pi + (-1)^n \frac{\pi}{3}$

C) $\theta = 2n\pi \pm \frac{\pi}{3}$

D) $\theta = 2n\pi + \frac{\pi}{3}$

3) $\int \frac{dx}{\sqrt{4-x^2}} =$

A) $\sin^{-1}\frac{x}{2} + C$

B) $\sin^{-1}\frac{x}{4} + C$

C) $\frac{1}{2}\sin^{-1}\frac{x}{2} + C$

D) $\frac{1}{2}\sin^{-1}x + C$

4) The number N of animals in a population at time t years is given by $N = 100 + Ae^{kt}$, for $A > 0$ and $k > 0$. Which of the following is the correct differential equation?

A) $\frac{dN}{dt} = k(N + 100)$

B) $\frac{dN}{dt} = -k(N - 100)$

C) $\frac{dN}{dt} = k(N - 100)$

D) $\frac{dN}{dt} = -k(N + 100)$

Answer Questions 5 – 7 on your own paper. Start each question on a new page.

- 5) a) Water is poured into a conical vessel of height 30 cm and radius 24 cm.
- i) Show that the volume of water is given by $V = \frac{16\pi h^3}{75}$, when the depth of water is h metres. (2)
 - ii) If the depth of water is increasing at the rate of $\frac{1}{2} \text{ cm/min}$, find the rate of increase of the volume of water when the depth of the water is 20 cm. (3)
- b) The rate of growth of the number of sheep on a farm is given by $\frac{dN}{dt} = k(N - 300)$, where N is the number of sheep and t is the time in years.
- i) Show that $N = 300 + Ae^{kt}$ is a solution to the differential equation. (1)
 - ii) If initially there are 400 sheep and two years later there are 700, find the number of sheep on the farm at the end of four years. (3)
- c) Find the general solution of $\sqrt{3} \sin \theta - \cos \theta = \sqrt{3}$ (3)
- 6) a) Find all values of x for which $2 \cos \left(3x + \frac{\pi}{6} \right) + \sqrt{3} = 0$ (3)
- b) Consider the function $f(x) = x^2 - 2x$
- i) Find the domain over which the function is monotonic increasing. (1)
 - ii) Find the inverse function $f^{-1}(x)$ over this restricted domain. (2)
 - iii) On the same set of axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ showing where the graphs intersect over their respective domains. (2)

- c) Find the area of the region bounded by the curve $y = \frac{1}{2} \sin^{-1} x$, the x axis and the line $x = 1$. (4)

- 7) a) Differentiate $\sin^{-1} 5x$ (1)

- b) Show that $\tan^{-1}(4) - \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$ (3)

- c) Consider the function $y = 3 \cos^{-1}\left(\frac{x}{2}\right)$

- i) Draw a neat sketch of the function, clearly showing the domain and range. (2)

- ii) Find the gradient of the tangent to the curve when $y = \pi$ (2)

- d) Differentiate $x \tan^{-1} x$ and hence evaluate $\int_0^1 \tan^{-1} x \, dx$ (4)

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS

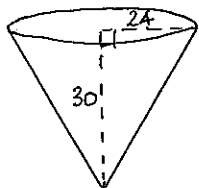
$$\begin{aligned}
 1) \tan^{-1}\left[\tan \frac{2\pi}{3}\right] &= \tan^{-1}(-\sqrt{3}) \\
 &= -\tan^{-1}(\sqrt{3}) \\
 &= -\frac{\pi}{3} \quad \textcircled{B}
 \end{aligned}$$

$$\begin{aligned}
 2) \cos \theta &= \frac{1}{2} \\
 \theta &= 2n\pi \pm \cos^{-1} \frac{1}{2} \\
 &= 2n\pi \pm \frac{\pi}{3} \quad \textcircled{C}
 \end{aligned}$$

$$3) \int \frac{dx}{\sqrt{4-x^2}} = \sin^{-1} \frac{x}{2} + C \quad \textcircled{A}$$

$$\begin{aligned}
 4) N &= 100 + Ae^{kt} \rightarrow Ae^{kt} = N - 100 \\
 \frac{dN}{dt} &= kAe^{kt} \\
 &= k(N - 100) \quad \textcircled{C}
 \end{aligned}$$

5) a)



$$\begin{aligned}
 i) \quad & \frac{r}{24} = \frac{h}{30} \\
 & r = \frac{24h}{30} \\
 & = \frac{4h}{5} \\
 \text{now } V &= \frac{1}{3}\pi r^2 h \\
 &= \frac{1}{3}\pi \cdot \frac{16h^2}{25} \cdot h \\
 &= \frac{16\pi h^3}{75}
 \end{aligned}$$

$$\begin{aligned}
 ii) \quad \frac{dV}{dh} &= \frac{48\pi h^2}{75} \\
 &= \frac{16\pi h^2}{25}
 \end{aligned}$$

$$\frac{dh}{dt} = \frac{1}{2}$$

$$\begin{aligned}
 \text{now } \frac{dV}{dt} &= \frac{dV}{dh} \cdot \frac{dh}{dt} \\
 &= \frac{16\pi h^2}{25} \cdot \frac{1}{2} \\
 &= \frac{8\pi h^2}{25}
 \end{aligned}$$

$$\begin{aligned}
 \text{When } h=20, \quad \frac{dV}{dt} &= \frac{8\pi \times 400}{25} \\
 &= 128\pi \text{ cm}^3/\text{min.}
 \end{aligned}$$

$$b) \frac{dN}{dt} = k(N - 300)$$

$$i) N = 300 + Ae^{kt}$$

$$\begin{aligned}
 \frac{dN}{dt} &= kAe^{kt} \\
 &= k(N - 300) \quad \therefore N = 300 + Ae^{kt} \text{ is a soln}
 \end{aligned}$$

$$ii) N = 300 + Ae^{kt}$$

$$\text{When } t=0, N=400$$

$$400 = 300 + A$$

$$A = 100$$

$$\therefore N = 300 + 100e^{kt}$$

$$\text{When } t=2, N=700$$

$$700 = 300 + 100e^{2k}$$

$$100e^{2k} = 400$$

$$e^{2k} = 4$$

$$2k = \ln 4$$

$$k = \frac{\ln 4}{2}$$

$$= 0.693$$

$$\therefore N = 300 + 100e^{0.693t}$$

$$\text{When } t=4, \quad N = 300 + 100e^{2.7t} \\ = 1899.058$$

$\therefore 1899$ sheep

$$c) \sqrt{3} \sin \theta - \cos \theta = \sqrt{3}$$

$$\sqrt{3} \sin \theta - \cos \theta = R \sin(\theta - \alpha) \\ = R \sin \theta \cos \alpha - R \cos \theta \sin \alpha$$

$$\begin{aligned} R \cos \alpha &= \sqrt{3} & \tan \alpha &= \frac{1}{\sqrt{3}} & R &= \sqrt{3+1} \\ R \sin \alpha &= 1 & \alpha &= \frac{\pi}{6} & &= 2 \end{aligned}$$

$$\therefore \sqrt{3} \sin \theta - \cos \theta = 2 \sin\left(\theta - \frac{\pi}{6}\right)$$

$$\text{i.e. } 2 \sin\left(\theta - \frac{\pi}{6}\right) = \sqrt{3}$$

$$\sin\left(\theta - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$\theta - \frac{\pi}{6} = n\pi + (-1)^n \frac{\pi}{3}$$

$$\theta = n\pi + (-1)^n \frac{\pi}{3} + \frac{\pi}{6}$$

$$\begin{aligned} \theta &= n\pi + \frac{\pi}{3} + \frac{\pi}{6} & \text{or} & & \theta &= n\pi - \frac{\pi}{3} + \frac{\pi}{6} \\ &= n\pi + \frac{\pi}{2} & & & &= n\pi - \frac{\pi}{6} \end{aligned}$$

$$b) a) 2 \cos\left(3x + \frac{\pi}{6}\right) + \sqrt{3} = 0$$

$$\cos\left(3x + \frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$$

$$3x + \frac{\pi}{6} = 2n\pi \pm \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$$

$$3x + \frac{\pi}{6} = 2n\pi \pm \frac{5\pi}{6}$$

$$3x = 2n\pi \pm \frac{5\pi}{6} - \frac{\pi}{6}$$

$$3x = 2n\pi + \frac{5\pi}{6} - \frac{\pi}{6} \quad \text{or} \quad 3x = 2n\pi - \frac{5\pi}{6} - \frac{\pi}{6}$$

$$3x = 2n\pi + \frac{2\pi}{3}$$

$$x = \frac{2n\pi}{3} + \frac{2\pi}{9}$$

$$3x = 2n\pi - \pi$$

$$x = \frac{2n\pi}{3} - \frac{\pi}{3}$$

$$b) f(x) = x^2 - 2x$$

$$i) f'(x) = 2x - 2$$

for monotonic increasing, $f'(x) > 0$

$$\text{i.e. } 2x - 2 > 0 \\ x > 1$$

$$ii) y = x^2 - 2x$$

$$x = y^2 - 2y$$

$$x+1 = y^2 - 2y + 1$$

$$x+1 = (y-1)^2$$

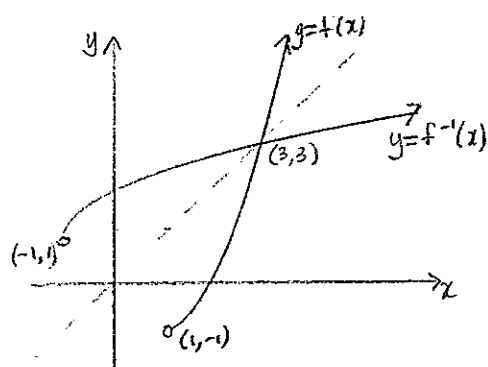
$$y-1 = \pm \sqrt{x+1}$$

$$y = 1 \pm \sqrt{x+1}$$

but $y > 1$

$$\therefore y = 1 + \sqrt{x+1}$$

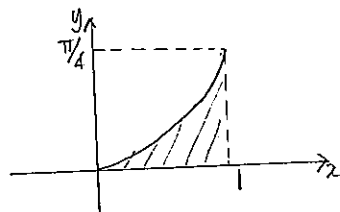
iii)



c) $y = \frac{1}{2} \sin^{-1} x$

$2y = \sin^{-1} x$

$\sin 2y = x$



$$A = \frac{\pi}{4} - \int_0^{\pi/4} \sin 2y \, dy$$

$$= \frac{\pi}{4} - \left[-\frac{1}{2} \cos 2y \right]_0^{\pi/4}$$

$$= \frac{\pi}{4} + \left[\frac{1}{2} \cos \frac{\pi}{2} - \frac{1}{2} \cos 0 \right]$$

$$= \frac{\pi}{4} + \left(0 - \frac{1}{2} \right)$$

$$= \left(\frac{\pi}{4} - \frac{1}{2} \right) \text{ sq units.}$$

7) a) $\frac{d}{dx} \sin^{-1} 5x = \frac{5}{\sqrt{1-25x^2}}$

b) $\tan^{-1}(4) - \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$

let $x = \tan^{-1}(4)$ and $y = \tan^{-1}\left(\frac{3}{5}\right)$

then $\tan x = 4$

then $\tan y = \frac{3}{5}$

now $\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$

$$= \frac{4 - \frac{3}{5}}{1 + 4 \cdot \frac{3}{5}}$$

$$= \frac{4 - \frac{3}{5}}{1 + \frac{12}{5}}$$

$$= \frac{\frac{20-3}{5}}{\frac{5+12}{5}}$$

$$= \frac{17}{17}$$

$$= 1$$

$\therefore \tan(x-y) = 1$

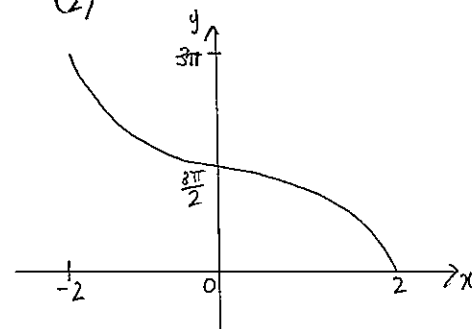
$$x-y = \tan^{-1} 1$$

$$= \frac{\pi}{4}$$

$\therefore \tan^{-1}(4) - \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$

c) $y = 3\cos^{-1}\left(\frac{x}{2}\right)$

i)



ii) $y = 3\cos^{-1}\left(\frac{x}{2}\right)$

$$y' = \frac{-3}{\sqrt{4-x^2}}$$

when $y = \pi$: $3\cos^{-1}\left(\frac{x}{2}\right) = \pi$

$$\cos^{-1}\left(\frac{x}{2}\right) = \frac{\pi}{3}$$

$$\frac{x}{2} = \frac{1}{2}$$

$$\therefore x = 1$$

$$\text{When } x=1: y' = \frac{-3}{\sqrt{4-1}}$$

$$= \frac{-3}{\sqrt{3}}$$

$$= -\sqrt{3}$$

$\therefore$ gradient is $-\sqrt{3}$.

$$\begin{aligned} \text{d) } \frac{d}{dx} [x \tan^{-1} x] &= \tan^{-1} x + x \cdot \frac{1}{1+x^2} \\ &= \tan^{-1} x + \frac{x}{1+x^2} \end{aligned}$$

$$\begin{aligned} \therefore \int_0^1 \tan^{-1} x + \frac{x}{1+x^2} dx &= [x \tan^{-1} x]_0^1 \\ \int_0^1 \tan^{-1} x dx &= [x \tan^{-1} x]_0^1 - \int_0^1 \frac{x}{1+x^2} dx \\ &= \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1 \\ &= \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) - (0 - 0) \\ &= \frac{\pi}{4} - \frac{1}{2} \ln 2 \end{aligned}$$